

Database Course

Ahmad Yoosofan

SQL 1

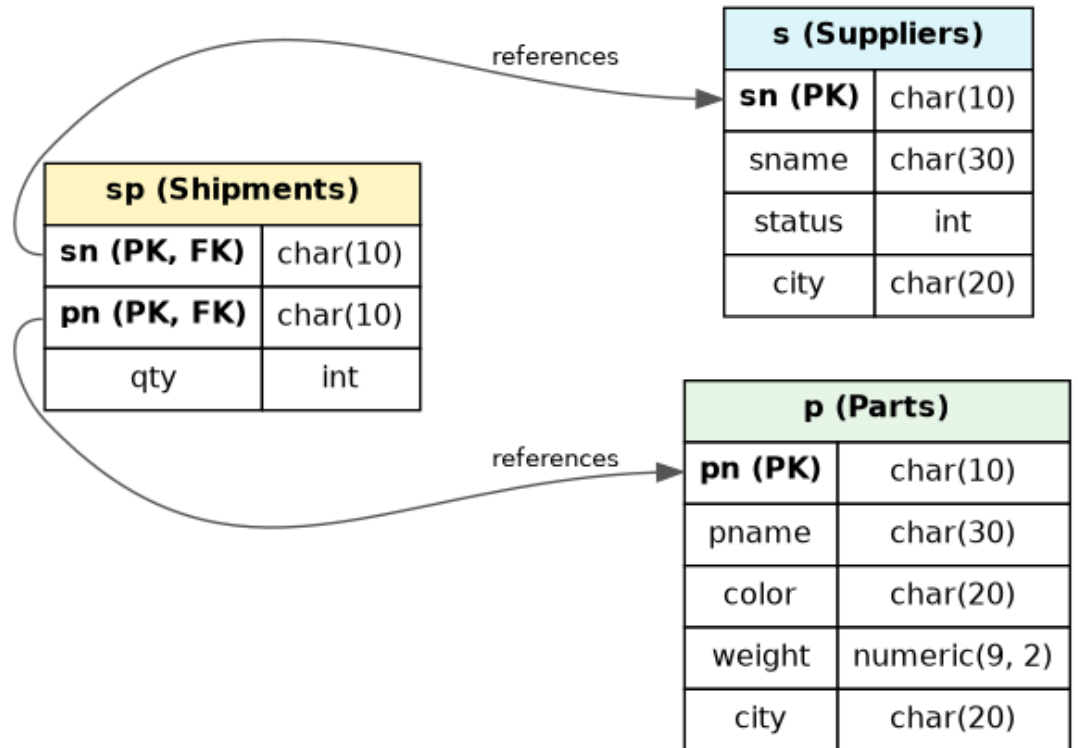
University of Kashan

<https://yoosofan.github.io>

<https://yoosofan.github.io/slide/db/sql1/>

Database Schema of SP

```
1 create table s (  
2     sn      char(10) primary key,  
3     sname   char(30),  
4     status  int default(0),  
5     city    char(20)  
6 );  
7  
8 create table p (  
9     pn      char(10) primary key,  
10    pname   char(30),  
11    color   char(20),  
12    weight  NUMERIC(9, 2),  
13    city    char(20)  
14 );  
15  
16 create table sp (  
17     sn      char(10) references s,  
18     pn      char(10) references p,  
19     qty     int default(0),  
20     primary key (sn, pn)  
21 );
```



DSL(Data Sub Language)

```
1 insert into s(sn, sname, status, city)
2 values('s1', 'Smith', 20, 'London')
3 ;
4
5 insert into p(pn, pname, color, weight, city)
6 values('p1', 'Nut' , 'Red' , 12.0, 'London')
7 ;
8
9 insert into p(pn, pname, color, weight, city)
10 values
11 ('p2', 'Bolt' , 'Green', 17.0, 'Paris' ),
12 ('p5', 'Cam' , 'Blue' , 12.0, 'Paris' ),
13 ('p6', 'Cog' , 'Red' , 19.0, 'London')
14 ;
```

```
delete sp;
```

```
drop table sp;
```

- DDL: Data Definition Language
- DML: Data Manipulation Language
- DCL: Data Control Language

1. DDL: create, drop
2. DML: insert, delete

```
1 insert into s(sn, sname, "status", city)
2 values('s4', 'Clark', 20, 'London')
3 ;
4 insert into s(sname, status, city, sn)
5 values('Adams', 30, 'Athens', 's5')
6 ;
7 insert into s
8 values('s6', 'Ali', 40, 'کاشان')
9 ;
```

DBMS(Database Management System)

- An application
- RDBMS (Relational Datababase Management System)
- SQL (Structured Query Language)
- DB2, Oracle, PostgreSQL, MySQL, SqlServer, MariaDB
- SQLite (Lack of DCL commands, each db on a file)

SQLite

1. [SQLite](#)
2. [SQLite browser](#)
3. [SQLite studio](#)
4. [SQLite-web](#)
5. [Database Viewer](#)

Online

1. [sql.js](#)
2. [sqlite online](#)
3. [sqlite-browser](#)
4. [sqlite-viewer](#)
5. [sqlfiddle](#)

نام قطعه‌ها را بیابید.

```
select pname  
from p  
;
```

```
p{pname};
```

pname
Nut
Bolt
Screw
Screw
Cam
Cog
Nut
Bolt

```

1 select sname, status
2 from s
3 ;

```

```

1 s{sname, status};
2
3 -- Relational Algebra

```

sname	status
Smith	20
Jones	10
Blake	30
Clark	20
Adams	30
Ali	40

```

1 select *
2 from s;
3 ;

```

sn	sname	status	city
s1	Smith	20	London
s2	Jones	10	Paris
s3	Blake	30	Paris
s4	Clark	20	London
s5	Adams	30	Athens
s6	Ali	40	کاشان

as (rename)

```
1 select sname, status + 4
2 from s;
3 ;
```

sname	status + 4
Smith	24
Jones	14
Blake	34
Clark	24
Adams	34
Ali	44

```
1 select sname, status + 4 as st4
2 from s;
3 ;
```

sname	st4
Smith	24
Jones	14
Blake	34
Clark	24
Adams	34
Ali	44

نام قطعه‌ها و وزن آنها را بیابید.

```
select pname, weight  
from p  
;
```

```
p{pname, weight} ;
```

pname	weight
Nut	12
Bolt	17
Screw	17
Screw	14
Cam	12
Cog	19
Nut	
Bolt	

NULL

```
1 insert into p(pn, pname, color, city)
2 values('p7', 'Nut', 'Red', 'London')
3 ;
```

```
1 insert into p(pn, pname, color, weight, city)
2 values('p8', 'Bolt', 'Green', null, 'Paris')
3 ;
```

1. null
2. not 'null'
3. not 0
4. not '0'
5. not "

```
1 create table s (
2   sn      char(10) primary key,
3   sname   char(30) not null,
4   status  int default 0,
5   city    char(20)
6 );
```

1. Do not know the value
2. Not applicable
 - Address: city, street, alley, number

نام قطعه‌ها و وزن آن‌ها را به گرم بیابید.

```
1 select pname, weight * 1000 as gweight
2 from p
3 ;
```

NULL * 1000 → NULL

pname	gweight
Nut	12000
Bolt	17000
Screw	17000
Screw	14000
Cam	12000
Cog	19000
Nut	
Bolt	

Where

```
1  -- نام عرضه کنندگان شهر کاشان را بیا بید
2
3  select sname
4  from s
5  where city = 'کاشان'
6  ;
```

```
-- (s where city = 'کاشان') {pname}
```

```
select sname
from s
where city = 'Paris'
;
```

sname
Jones
Blake

- Arithmetic Operators
 - `+ - \* / %`
- Comparison Operators
 - `= < > >= <= <>`

شماره قطعه‌های عرضه شده را بیابید.

```
select pn  
from sp  
;
```

pn
p1
p2
p3
p4
p5
p6
p1
p2
p2
p2
p4
p5
p2

نام قطعه‌های عرضه شده را بیابید.

```
select pname  
from p, sp  
where p.pn = sp.pn  
;
```

```
(  
  (  
    (  
      p rename pn as ppn  
    )  
    times sp  
  ) where ppn = pn  
) {pname}
```

pname
Nut
Bolt
Screw
Screw
Cam
Cog
Nut
Bolt
Bolt
Bolt
Screw
Cam
Bolt

join

نام قطعه‌های عرضه شده را بیابید.

```
(p join sp) {pname}
```

```
select pname  
from p join sp on p.pn=sp.pn  
;
```

```
select pname  
from p natural join sp  
;
```

```
select pname  
from p join sp using(pn)  
;
```

نام قطعه‌هایی را بیابید که در شهر آن قطعه‌ها عرضه کننده‌ای وجود داشته باشد

```
select pname  
from p join s using(city)  
;
```

```
select pname  
from p natural join s  
;
```

```
select distinct pname  
from p natural join s  
;
```

pname
Nut
Nut
Bolt
Bolt
Screw
Screw
Cam
Cam
Cog
Cog
Nut
Nut
Bolt
Bolt

pname
Nut
Bolt
Screw
Cam
Cog

نام قطعاتی را بیابید که عرضه‌کننده‌ای از شهر کاشان آنها را عرضه کرده باشد.

```
select pname
from (p natural join sp)
join s on s.sn=sp.sn
where s.city = 'کاشان'
;
```

```
select pname
from (p natural join sp)
join s using(sn)
where s.city = 'کاشان'
;
```

pname
Bolt

SQLite (I)

1. sqlite3
2. sqlite3 sp.sqlite
3. .exit or .quit
4. .help
5. .read sp.sql
6. .output sp2.sql
7. .dump
8. .output

1. .open sp.sqlite
2. .databases
3. .backup FILE
4. .restore FILE
5. .system CMD
6. .system clear
7. .tables
8. .schema s

Use Another name for a Table in Query

```
create table t (  
  a int primary key,  
  name char(20)  
);  
  
insert into t values (1, 'a'), (2, 'b');
```

```
select *  
from t, t as M;
```

a	name	a	name
1	a	1	a
1	a	2	b
2	b	1	a
2	b	2	b

```
select t.name  
from t, t as M  
where t.a < M.a;
```

name
a

```
select *  
from t join t as M  
on t.a < M.a;
```

a	name	a	name
1	a	2	b

نام قطعاتی را بیابید که وزن آنها دست کم از وزن یک قطعه دیگر باشد نام تکراری در پاسخ نیاید.

نام همه قطعات را بیابید به جز قطعه یا قطعه‌هایی که کمترین وزن را دارند

```
1 select T.pname
2 from p as T
3 ;
```

```
1 select T.pname
2 from p as T, p
3 where p.weight < T.weight
4 ;
```

```
1 select distinct T.pname
2 from p as T, p
3 where p.weight < T.weight
4 ;
```

```
1 select distinct T.pname
2 from p as T join p on
3   p.weight < T.weight
4 ;
```

pname
Bolt
Screw
Cog

نام قطعه‌های عرضه شده را همراه با نام عرضه‌کنندگان‌شان بیابید زوج نام تکراری در پاسخ نیاید.

```
select pname, sname
from s, sp, p
where s.sn = sp.sn and
      p.pn = sp.pn
;
```

```
select distinct pname, sname
from s natural join sp
join p using(pn)
;
```

pname	sname
Nut	Smith
Bolt	Smith
Screw	Smith
Cam	Smith
Cog	Smith
Nut	Jones
Bolt	Jones
Bolt	Blake
Bolt	Clark
Screw	Clark
Cam	Clark
Bolt	Ali

نام قطعاتی را بیابید که وزنشان دست کم از وزن یک قطعه با رنگ قرمز کمتر باشد

```
select distinct T.pname
from p as T, p
where p.weight > T.weight
and p.color='Red'
;
```

```
select distinct T.pname
from p as T join p on
p.weight > T.weight
where p.color='Red'
;
```

```
select distinct p.pname
from p as p1 join p on
p1.weight > p.weight and
p1.color = 'Red'
;
```

pname
Nut
Bolt
Screw
Cam

LIKE

نام شهرهای قطعاتی را بیابید که با P آغاز شده باشد

```
select city
from p
where city like 'P%'
;
```

نام قطعاتی را بیابید که کاراکتر دوم نامشان o باشد.

```
select pname
from p
where city like '_o%'
;
```

نام شهر قطعاتی را بیابید که درون نام شهر آنها رشته is وجود داشته باشد

```
select city
from p
where city like '%is%'
;
```

نام قطعات و شهرهای آنها را بیابید که شهر آنها دست کم سه حرفی باشند و با رشته bn_ آغاز شود.

```
select pname, city
from p
where city like 'bn\_%'
;
```


escape

```
select pname
from p
where city like 'P\_%' escape '\'
;
```

```
select pname
from p
where city like 'P!_%' escape '!'
;
```

```
select pname
from p
where city like 'P#_%' escape '#'
;
```

```
select pname
from p
where city like "an\_%" escape "\"
; -- "
```

ORDER BY

نام قطعاتی را بیابید که در شهر پاریس باشند و پاسخ بر پایه نام قطعه از کوچک به بزرگ مرتب شده باشد.

```
select pname
from p
where city='Paris'
order by pname
;
```

نام و وزن قطعاتی را بیابید که در شهر پاریس هستند و پاسخ بر پایه وزن قطعه از کوچک به بزرگ مرتب شده باشد

```
select pname, weight
from p
where city='Paris'
order by weight
;
```

```
select pname, weight
from p
where city='Paris'
order by weight asc
;
```

pname	weight
Bolt	NULL
Cam	12
Bolt	17

نام و وزن قطعاتی را بیابید که در شهر پاریس هستند و پاسخ بر پایه وزن قطعه از بزرگ به کوچک مرتب شده باشد

```
select pname, weight
from p
where city='Paris'
order by weight desc
;
```

pname	weight
Bolt	17
Cam	12
Bolt	

```
select pname, weight
from p
where city='Paris' and weight is not null
order by weight desc
;
```

pname	weight
Bolt	17
Cam	12

BETWEEN

نام و وزن قطعاتی را بیابید که وزن شان بین ۱۲ و ۱۴ باشد

pname	weight
Nut	12
Screw	14
Cam	12

```
select pname, weight
from p
where weight >= 12 and weight <= 14
;
```

```
select pname, weight
from p
where weight between 12 and 14;
```

نام و وزن قطعاتی را بیابید که وزن شان بین ۱۲ و ۱۴ نباشد

pname	weight
Bolt	17
Screw	17
Cog	19

```
select pname, weight
from p
where not (weight >= 12 and weight <= 14)
;
```

```
select pname, weight
from p
where weight not between 12 and 14
;
```

```
select pname, weight
from p
where weight < 12 or weight > 14
;
```

نام قطعاتی را بیاید که عرضه کنندهای در شهر آن قطعه‌ها آنها را عرضه کرده باشد

Record Comparison

```
select pname
from p, s, sp
where p.city = s.city and
      p.pn = sp.pn and
      s.sn = sp.sn
;
```

```
select pname
from p, s, sp
where (p.city, p.pn) = (s.city, sp.pn)
      and s.sn = sp.sn
;
```

```
select pname
from p join s on
      p.city = s.city
join sp on
      (p.pn, s.sn) = (sp.pn, sp.sn)
;
```

```
select pname
from p natural join sp natural join s
;
```

Union

نام قطعاتی از شهر پاریس را بیابید که وزن آنها بیشتر از ۱۲ است.

```
select distinct pname
from p
where city = 'Paris' or
       weight > 12;
```

```
select pname
from p
where city='Paris'
union
select pname
from p
where weight>12;
```

pname
Bolt
Cam
Cog
Screw

```
select pname
from p
where city = 'kashan'
union all
select pname
from p
where weight>10
;
```

pname
Nut
Bolt
Screw
Screw
Cam
Cog

Style of Writing

```
select pname
from p
where city='Paris'
union
select pname
from p
where weight>12
;
```

```
select pname
from p
where city='kashan'
union
select pname
from p
where weight>10
;
```

```
select pname
from p
where city='kashan'

union

select pname
from p
where weight>10
;
```

Intersect

```
select pname
from p
where city='Paris'
intersect
select pname
from p
where weight>10
;
```

```
select distinct pname
from p
where city='Paris' and
weight>10
;
```

pname
Bolt
Cam

```
select pname
from p
where city = 'Paris'
intersect all -- sqlite error
select pname
from p
where weight > 10
;
```

```
select pname
from p
where city='Paris' and
weight>10
;
```

pname
Bolt
Cam

Except

```
select pname
from p
where city = 'Paris'
except
select pname
from p
where weight > 14
;
```

```
select distinct pname
from p
where city='Paris' and
weight<=14
;
```

pname
Cam

```
select pname
from p
where city='Paris'
except all -- sqlite error
select pname
from p
where weight>10
;
```

```
select pname
from p
where city='Paris' and
weight<=14
;
```

pname
Cam

نام شهرهای قطعاتی را بیابید که در آنها عرضه‌کننده‌ای وجود ندارد

```
select city
from p
except
select city
from s
;
```

city
Oslo

شماره قطعات و شماره عرضه کنندگانی را بیابید که قطعات یاد شده را آن عرضه کنندگان عرضه نکرده باشند

```
1 select pn, sn
2 from p, s
3 except
4 select pn, sn
5 from sp;
```

```
1 select distinct p.pn, s.sn
2 from p, s, sp -- incorrect
3 where (s.sn, p.pn) <> (sp.sn, sp.pn)
4 ; -- s1, p2
```

```
1 select * from (
2     select pn, sn
3     from p, s
4     except
5     select pn, sn
6     from sp
7 ) order by pn,sn;
```

```
1 select distinct p.pn, s.sn
2 from p, s, sp -- incorrect
3 where (s.sn, p.pn) <> (sp.sn, sp.pn)
4 order by sp.pn,sp.sn
5 ; -- s1, p2
```

```
1 select pn,sn
2 from sp order by pn,sn
3 ; -- s1, p2
```

نام قطعات و نام عرضه کنندگانی را بیابید که قطعات یاد شده را آن عرضه کنندگان عرضه نکرده باشند

```
select pname, sname -- نادرست
from p, s
except
select pname, sname
from p natural join sp
natural join s;
```

```
select pname, sname from p, s
except
select pname, sname
from s natural join sp
join p using(pn);
```

```
select sname , pname
from (
  select pn, sn from p, s
  except
  select pn, sn from sp
) join p using (pn)
join s using (sn);
```

pname	sname	pname	sname
		Cog	Clark
Bolt	Adams	Cog	Jones
Cam	Adams	Nut	Adams
Cam	Ali	Nut	Ali
Cam	Blake	Nut	Blake
Cam	Jones	Nut	Clark
Cog	Adams	Screw	Adams
Cog	Ali	Screw	Ali
Cog	Blake	Screw	Blake
		Screw	Jones

زوج نام عرضه کنندگانی را بیابید که در یک شهر باشند و پاسخ تکراری نداشته باشید

```
-- (1) نادرست
select s.sname, T.sname
from s, s as T
where s.city = T.city
;
```

```
-- (2) نادرست
select s.sname, T.sname
from s, s as T
where s.city = T.city and
      s.sn != T.sn
;
```

```
-- (3)
select s.sname, T.sname
from s, s as T
where s.city = T.city and
      s.sn < T.sn
;
```

```
-- (4)
select s.sname, T.sname
from s as T join s using(city)
where s.sn < T.sn
;
```

```
-- (5)
select s.sname, T.sname
from s as T join s on
      T.city = s.city and
      s.sn < T.sn
;
```

sname	sname
Smith	Clark
Jones	Blake

LIMIT

```
select distinct city  
from p  
order by weight, city  
;
```

city
London
Oslo
Paris

```
select distinct city  
from p  
order by weight, city  
limit 2  
;
```

city
London
Oslo

Scalar value(I)

شماره و وزن قطعاتی را بیابید که کمترین وزن را داشته باشند.

```
1 select pn, weight -- incorrect
2 from p
3 order by weight asc
4 limit 1
5 ;
```

pn	weight
P7	NULL

```
1 select pn, weight
2 from p
3 where weight is not null
4 order by weight asc
5 limit 1
6 ;
```

pn	weight
p1	12

```
1 select weight
2 from p
3 where weight is not null
4 order by weight asc
5 limit 1
6 ;
```

weight
12

شماره و وزن قطعاتی را بیابید که کمترین وزن را داشته باشند.

```
select pn, weight
from p    -- incorrect
where weight = (
    select weight
    from p
    order by weight asc
    limit 1
);
```

```
select pn, weight
from p
where weight = (
    select weight
    from p
    where weight is not null
    order by weight asc
    limit 1
);
```

```
1 select pn, 1 as qt
2 from p
3 where city = 'Paris'
4 ;
```

pn	qt
p2	1
p5	1
p8	1


```
1 select pn, (  
2     select weight  
3     from p  
4     where weight is not null  
5     order by weight asc  
6     limit 1  
7 ) as qt  
8 from p  
9 where city = 'Paris' ;
```

pn	qt
p2	12
p5	12
p8	12

Update(I)

```
update p
set weight = weight + 15
where city = 'Oslo'
;
```

```
update s
set status = status * 2,
    city = 'kashan'
where city = 'London'
;
```

```
update employees
set email = LOWER(
    firstname || "." ||
    lastname || "@chinookcorp.com"
);
```

```
update P
set weight = null
where pn='p6'
;
```

Update(II)

```
update s
set
  status = case
    when city = 'london' then status * 2
    else status
  end;
```

```
update s
set
  status = case
    when city = 'London' then status * 2
    when city = 'Paris'  then status * 3
    else status
  end;
```

```
update s
set
  status = case
    when city = 'London' then status / 4
    when city = 'Paris'  then status / 3
    else status
  end;
```

```
update s
set city = 'kashan',
    status = case
      when status > 20 then 53
      else status
    end
where sname = 'Smith'
returning sn, sname;
```

Update or Replace

```
1 CREATE TABLE users (  
2     id INTEGER PRIMARY KEY,  
3     email CHAR(20) UNIQUE,  
4     name CHAR(15)  
5 );
```

```
1 INSERT INTO users VALUES  
2     (1, 'ada@x.com', 'Ada'),  
3     (2, 'boris@x.com', 'Boris');
```

```
1 -- This would fail with a UNIQUE  
2 -- constraint violation:  
3  
4 UPDATE users SET email = 'boris@x.com'  
5 WHERE id = 1;
```

```
1 -- OR REPLACE deletes the conflicting r  
2 -- then performs the update.  
3 UPDATE OR REPLACE users  
4 SET email = 'boris@x.com'  
5 WHERE id = 1;
```

```
1 SELECT * FROM users;-
```

id	email	name
1	boris@x.com	ada

Transaction

Data Types

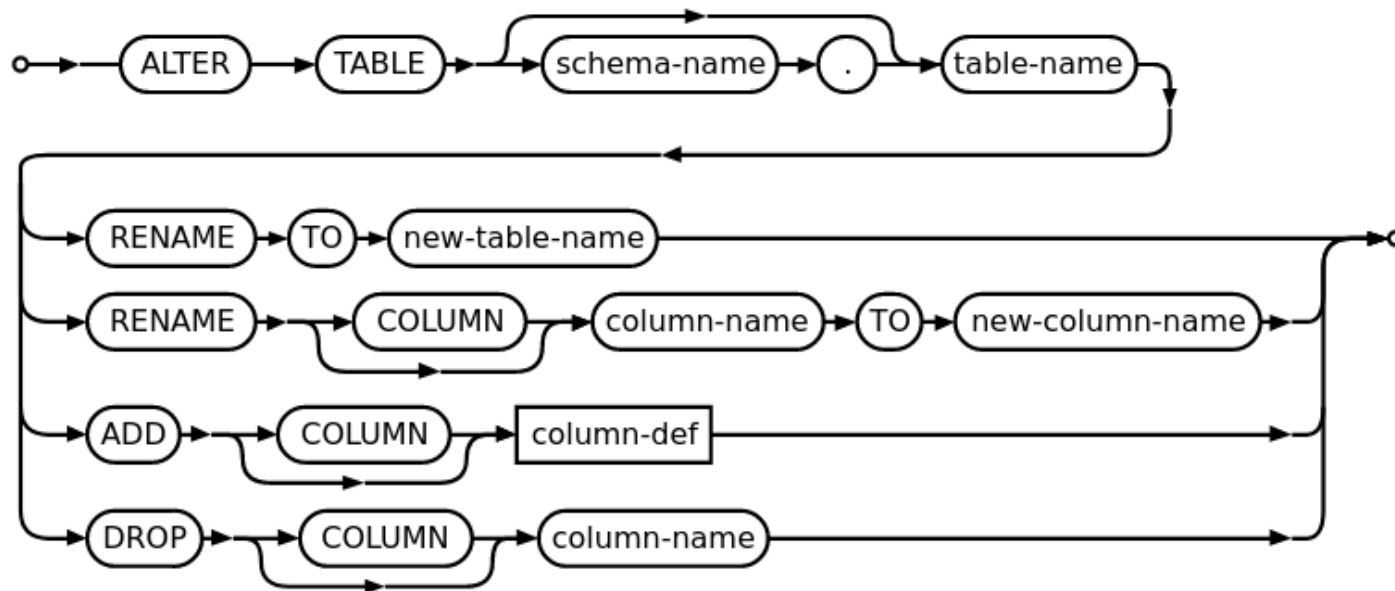
Alter Table

DDL

```
alter table sp add "comment" varchar(50);
```

```
alter table sp drop "comment";
```

```
alter table sp add "comment" varchar(50) default ' ';
```



END

